

Homework 6

Due Date: September 11, 2019 by 5:00pm

MAT 022A Linear Algebra

Instructor: Mikhail Gaerlan

Instructions: Write solutions neatly and show all work. No need to show all steps when row reducing. Box your solutions, and explain your answers if necessary. Upload a PDF file of your assignment to Gradescope. When uploading to Gradescope, make sure to select which pages correspond to which question.

1. (Strang 6.4.7) Find an orthogonal matrix Q that diagonalizes this symmetric matrix:

$$S = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -2 \\ 2 & -2 & 0 \end{bmatrix}$$

2. (Hill 7.2.1) Find a least squares solution to $A\mathbf{x} = \mathbf{b}$.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \\ -1 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix}$$

3. (Strang 4.2.11) Project $\mathbf{b}$ onto the column space of A by solving $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ and $\mathbf{p} = A\hat{\mathbf{x}}$:

$$\text{a. } A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \quad \text{b. } A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$$

4. (Strang 7.2.1) Find the eigenvalues of these matrices. Then find singular values from $A^T A$:

$$A = \begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix}$$

For each A , construct V from the eigenvectors of $A^T A$ and U from the eigenvectors of AA^T . Check that $A = U\Sigma V^T$.

5. (Strang 7.2.2) Find $A^T A$ and V and Σ and $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$, and the full SVD:

$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = U\Sigma V^T$$